



Gel DNA Extraction

Designing Life with AI

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Description

Gel extraction is used when PCR produces multiple bands, allowing us to isolate only the correct fragment. However, this method is slower and provides a much lower yield than the standard PCR clean-up protocol.

Steps

DNA extraction from the gel, based on the Macherey-Nagel Gel Extraction Kit from Vazyme:

<https://www.vazymeglobal.com/product-center/dna-clean-up/fastpure-gel-dna-extraction-mini-kit>

1. Slice the gel with a scalpel to retrieve the correct band only (under UV light).
2. Transfer the gel slice to a pre-weighed tube and record the weight.
3. Add the recommended volume of binding buffer (based on gel weight) to dissolve the agarose.
4. Incubate at $\sim 50^{\circ}\text{C}$ until the gel slice is fully melted, vortexing occasionally.
5. Load the dissolved gel solution onto the provided spin column.
6. Centrifuge according to the kit instructions and discard flow-through.
7. Wash the column with the kit wash buffer (typically 2 washes).
8. Dry-spin to remove residual ethanol.
9. Elute DNA in 11 μl MilliQ water.

Protocol from Vazyme website (Gel extraction program)

1. After DNA electrophoresis for fractionating DNA fragments, excise the single stripe of DNA fragment from the agarose gel under UV light. It is recommended to use absorbent paper to absorb apparent moisture of gel and minimize the size of the gel slice by removing extra agarose as possible as you can. Weigh the gel slice (without microcentrifuge tube) to calculate its volume: the volume of 100 mg gel slice is approximately 100 μl , assuming the density is 1 g/ml.
2. Add equal volume Buffer GDP, incubate at $50\text{--}55^{\circ}\text{C}$ for 7–10 min (according to the gel size, adjust incubation time until the gel completely dissolved). Invert the tube 2 times during the incubation. *(Addition of 1–3 volumes of Buffer GDP will not influence DNA recovery efficiency. If the DNA fragment to be recovered < 100 bp, 3 volumes of Buffer GDP need to be added; when the gel slice has dissolved completely, add 1 volume of isopropanol and mix thoroughly, then continue to the next step.)*
3. Centrifuge briefly to bring the sample to the bottom of the tube, insert the FastPure DNA Mini Columns-G into the Collection Tubes 2 ml, carefully transfer the solution maximally of 700 μl once a time to the filtration columns, centrifuge at 12,000 rpm ($13,800 \times g$) for 30–60 sec.
4. Discard the filtrate and add 300 μl of Buffer GDP to the column, incubate at room temperature for 1 min, centrifuge at 12,000 rpm ($13,800 \times g$) for 30–60 sec.
5. Discard the filtrate and add 700 μl of Buffer GW (check if absolute ethanol has been added in advance!) to the column, centrifuge at 12,000 rpm ($13,800 \times g$) for 30–60 sec. *(Please add Buffer GW around the adsorption column wall, or add Buffer GW back cover and mix it upside down for 2–3 times to help completely flush the salt adhering to the tube wall.)*

6. Repeat step 5. (*Flushing with Buffer GW twice can ensure that the salt is completely removed and eliminate the impact on subsequent experiments.*)
7. Discard the filtrate and centrifuge the empty column at 12,000 rpm ($13,800 \times g$) for 2 min.
8. Insert the column into a clean 1.5 ml microcentrifuge tube, add 20–30 μ l of Elution Buffer to the center of the column membrane, incubate for 2 min, and then centrifuge at 12,000 rpm ($13,800 \times g$) for 1 min. Discard the column, store the obtained DNA at -20°C . (*Transferring the supernatant of step 8 to the column to elute again and preheating the Elution Buffer to 55°C (when DNA fragment >3 kb) may be helpful to increase the recovery efficiency.*)